Customer Segmentation: Data-Driven Research on Transaction-Level Data of JD.com

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Abstract

As a premier e-commerce platform in China, JD.com offers significant research potential through its extensive user data, clicks, and transaction records. This study utilizes the JD transaction-level dataset from the *2020 MSOM Data Driven Research Challenge* to predict consumer preferences and extract data-driven insights into customer behavior. Following the initial adoption of the k-means clustering methodology, a manual classification-based approach was subsequently proposed to mitigate the issue of boundary blurring. Various machine learning models, including Logistic Regression, Decision Tree, Random Forests, and Stacking, were subsequently employed to enhance the predictive analysis of users' Stock Keeping Unit (SKU) preferences. The implemented model achieved an accuracy rate of 74%, thereby advancing the comprehension of JD's heterogeneous customer base. This study presents a novel perspective for optimizing user experience on e-commerce platforms and contributes to the sustainable development of the industry.

CCS Concepts

• **Computing methodologies** → Machine learning; Machine learning approaches.

Keywords

JD.com, Customer segmentation, User classification, Machine learning

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1 INTRODUCTION

In recent years, the rapid advancement of e-commerce platforms has rendered online shopping an indispensable aspect of daily life [1]. JD.com, as a leading e-commerce platform in China, has amassed extensive user purchase data and transaction records, facilitated by its vast customer base and efficient logistics system [2]. This rich dataset includes not only user purchasing behavior but also multidimensional information such as product and user characteristics, offering valuable opportunities for a profound understanding of user behavior and the optimization of customer segmentation strategies (Ibid). Despite extensive research on user behavior within e-commerce platforms, most studies tend to focus on specific transaction factors or short-term behavior predictions, lacking a comprehensive understanding and universal analysis of users' overall shopping behavior. Furthermore, existing segmentation efforts predominantly rely on single data sources or specific features, such as click-through rates and browsing history, thereby overlooking the benefits of multidimensional data fusion [3]. Consequently, the effective utilization of big data technology to analyze user behavior from multiple perspectives and enhance the accuracy and personalization of customer segmentation remains an urgent challenge that necessitates resolution.

This study aims to address the identified research gap by utilizing JD's transaction-level data from the *2020 MSOM Data Driven Research Challenge* as the primary dataset. Through rigorous data visualization techniques, this research seeks to uncover prevalent patterns within JD's transaction data. The article not only performs an in-depth analysis of consumer purchasing behaviors but also employs clustering methodologies to categorize user groups based on the characteristics of products they clicked, thereby elucidating the shopping preferences of distinct user segments. Additionally, this paper applies regression models to predict users' purchasing power levels, uses diverse classification models to predict users' consumption preferences, and applies feature engineering and parameter tuning to optimize model performance. By leveraging these innovative approaches, this study provides data-driven insights into customer behavior and strategic marketing recommendations for the JD platform. Furthermore, it introduces new perspectives and methodologies for user behavior analysis and customer segmentation within the e-commerce sector. The ultimate objective is to enhance customer satisfaction and user experience on e-commerce platforms, thereby promoting the sustainable development of the industry as a whole.

2 GENERAL DESCRIPTION

2.1 Dataset Introduction

The dataset used in this study was JD's transaction-level data during March 2018, compiled by the *2020 MSOM Data-driven Research Challenge* [2]. It includes information on over 2.5 million customers (457,298 made purchases) and 31,868 SKUs from one product category during the month.

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Table 1: List of variable information.

Variable	Data type	Description	Sample Value
sku_ID	string	Unique identifier of a product	B4822497a5
Type	int	First-party owned (1P) or third-party owned (3P) SKU	1
brand_ID	string	Brand unique	C840ce7809
attribute1	int	First key attribute of the category	3
attribute2	int	Second key attribute of the category	60
user_ID	string	User unique identification code	000000f736
user_level	int	User level	10
plus	string	If user has a PLUS membership	0
gender	int	User gender (F: female; M: male; U: unknown)	F
age	string	User age range (≤15; 16–25; 26–35; 36–45; 46–55; ≥56; U: unknown)	26-35
marital_status	string	User marital status (M: married; S: single; U: unknown)	M
education	int	User education level (1: less than high school; 2: high school diploma or equivalent; 3: bachelor's degree; 4: postgraduate degree; −1: unknown)	3
city_level	int	City level of user address (ranging between 1 and 5: level 1 corresponds to highly industrialized cities; level 2 cities correspond to provincial capitals; level 3–5 cities are smaller cities; if there are no data, then the value is −1)	1
purchase_power	int	User purchase power (user's estimated purchase power, ranging from 1 to 5 with 1 being the highest purchase power; −1 if there is no estimation)	2
order_ID	string	Order unique identification code	3b76bfcd3b
original_unit_price	float	Original list price	99.9
final_unit_price	float	Final purchase price	53.9
order_time	string	Specific time at which the order gets placed (format: yyyy-mm-dd HH:MM:SS)	2018-03-01 11:10:40.0

Through data organization and variable merging, this article first lists the basic information of variables required for subsequent data exploration, as shown in Table 1.

2.2 Visual Information

By merging effective variables, potential relationships between data can be discovered. For example, Figure 1 below compares the transportation time of self-operated products (1P) and third-party products (3P).

In Figure 1, the labels Type1 and Type2 on the horizontal axis denote self-operated and non-self-operated products, respectively, while the vertical axis represents the time interval from order placement to item receipt. The data illustrate that the median delivery time for 1P products is comparatively short, suggesting that the delivery duration for the majority of self-operated items is faster relative to non-self-operated products. The brevity of the box for 1P products indicates a relatively concentrated delivery time distribution with minimal variability. This implies that 3P goods may

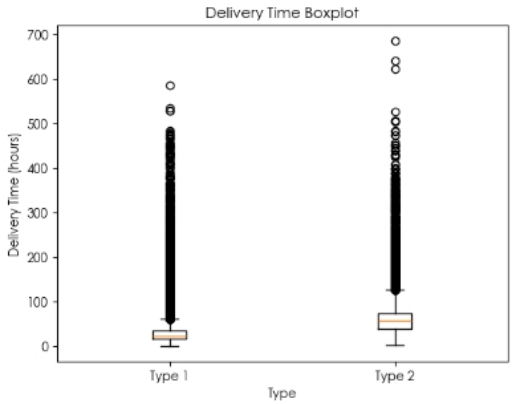

Figure 1: Box plot of transportation time for different types of products.

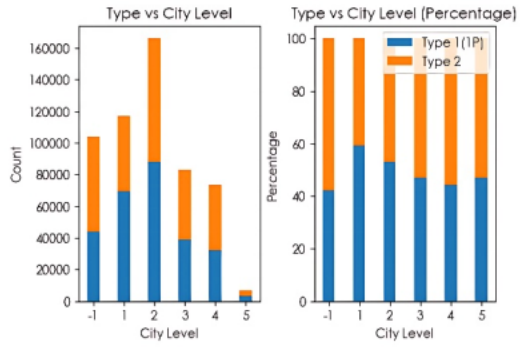

Figure 2: Histogram of SKU type and city level.

encounter increased delivery time uncertainty, attributable to factors such as supply chain complexities and inventory fluctuations.

The research also compares the selection proportion of 1P or 3P goods on city_level by merging the user and order information. In Figure 2, it is evident that city levels 1 and 2 exhibit higher order counts compared to levels 3, 4, and 5. Additionally, there is a notable trend indicating that residents in more prosperous cities are increasingly opting for 1P SKUs. The percentage distribution reinforces this observation, as higher city levels show a greater preference for Type 1 products.

2.3 Methods

The study classified variables into numerical and categorical types, reclassified multiple variables through data cleaning and variable preprocessing, and generated additional variables for feature selection.

2.3.1 Data Preprocessing. (a) Data cleaning and organization

The research addressed the issue of missing and anomalous values, such as negative user age values, by eliminating entire samples (rows) that encompass meaningless or unknown data. The study sought to prevent duplicative categorization by removing redundant SKUs. Additionally, the paper re-encoded particular data types to facilitate subsequent model processing. For instance, the user's gender was encoded, with "0" representing female and "1" representing male. The age range was also segmented and encoded, assigning "0" to individuals aged ≤ 15 years, "1" to those aged 16-25 years, "2" to those aged 26-35 years, "3" to those aged 36-45 years, and "4" to those aged ≥ 56 years. Furthermore, concerning the processing of temporal data within variables, these data were reformatted into a standard form to enable the calculation of time intervals.

(b) Interval scaling of numerical features

To facilitate later classification, the paper scaled attributes using 'Min-Max Scaling' or 'Z-score Normalization'. The Min-Max Scaling method was used to perform linear transformation on the variables Minimum merged attributes, Maximum merged attributes, and average discount, mapping their values to the interval [0, 1] [4]. The formula for Min-Max Scaling is as follows.

$$X' = \frac{X - X_{\min}}{X_{\max} - X_{min}} \tag{1}$$

Z-score Normalization method was used for original and final unit prices, with data outside the threshold being removed [5]. The formula for Z score Normalization is as follows, where μ is mean and σ is standard variance.

$$Z = \frac{X - \mu}{\sigma} \tag{2}$$

(c) New attributes

To evaluate the sku_category from numerous aspects, the paper took more than mere discount into account, introducing the impact of final_unit_price on avg_disount. It tried to ensure the weight of each part by calculating the contribution it does to the total variance. Hence, new attributes were created to evaluate SKU categories more comprehensively.

The calculations are shown as follows:

$$discount = \frac{\text{original_unit_price} - final_unit_price}{original_unit_price} \tag{3}$$

It is worth noting that the discount data calculated in this way means that the smaller the value, the less discount the product receives (not much different from the original price).

Hence,

$$\begin{aligned} \text{average}_discount \quad = \quad & discount * weight_discount \\ & + final_unit_price * weight_final_price \end{aligned} \tag{4}$$

In addition, in order to aggregate the effects of attributes 1 and 2, the article added the two attribute values of the corresponding product and used the Min-Max Rescaling to obtain the final attribute data of the product, denoted as std_combined_attribute. The higher the value, the better the quality of the product.

(d) Cluster – SKU classification

To classify users effectively, the paper first categorized SKU_IDs. Analyzing the inertia for various k values reveals that four categories can finally be obtained — luxury, cost-effective, bargain, and others — balanced both category sizes and information retention.

The study explored both machine learning (k-means) and manual classification methods, ultimately opting for the manual approach due to its superior explanatory power and balanced category sizes.

① Machine learning classification

Using preprocessed variable including avg_discount and std_combined_attribute, the paper applied k-means to categorize the SKUs. However, this method resulted in imbalanced categories, with a disproportionate number of luxury and others, impacting subsequent classifications.

② Manual classification

In light of the imbalanced k-means results, the study tried to manually categorized the SKUs into four groups based on quartiles of discounts and attributes. This method yielded distinct and balanced categories, particularly for bargain and luxury items.

To explain, SKUs were categorized as "Luxury products" when the avg_discount was ≤ 25% and the std_combined_attribute was > 75%. Those with an avg_discount > 25% and std_combined_attribute > 50% were labeled as "Cost-effective products". SKUs with an avg_discount > 75% and std_combined_attribute ≤ 25% fell into the "Bargains" category. All other SKUs were classified as "Others".

2.3.2 Model Introduction. This research introduces a range of machine learning models for predicting and classifying user behavior.

Table 2: Regression results of the two models.

Regression method	Train MSE	Test MSE
Linear Regression	0.240	0.241
Decision Tree Regression	0.234	0.240

Among these, the Logistic regression model, introduced by David Cox in 1958, primarily addresses binary classification problems [6]. Its notable advantages include the capacity to predict event probabilities and yield interpretable results, particularly for linearly separable data (Ibid). In this study, Logistic regression models were leveraged to forecast users' purchasing preferences, thereby aiding JD's recommendation system in providing more precise product suggestions. The Decision Tree model, initially proposed by John Ross Quinlan in the 1970s, is applicable to both classification and regression tasks [7]. This model excels in handling nonlinear relationships and is lauded for its ease of understanding and visualization, making it particularly suitable for small to medium-sized datasets (Ibid). Furthermore, the Random Forest model, an ensemble learning technique introduced by Leo Breiman in 2001, mitigates the error and uncertainty associated with individual models by aggregating the outcomes of multiple decision trees (Ibid). Additionally, this study incorporates the Stacking model, which enhances model performance by amalgamating the predictive results of multiple base learners, thereby rendering the final predictions more accurate and reliable [8].

2.3.3 Feature Engineering and Parameter Tuning. In the domain of data preprocessing and model training, feature engineering and parameter tuning are critical components. This study introduces Recursive Feature Elimination (RFE) as an advanced feature selection technique aimed at enhancing model performance by systematically removing non-essential features [9]. Regarding parameter tuning, this article examines two methodologies: Grid Search and Randomized Search. Grid Search is characterized by exhaustively exploring all potential parameter combinations to identify the optimal global solution [10]. Conversely, Randomized Search operates by arbitrarily selecting parameter combinations within the specified parameter space (Ibid). Compared to Grid Search, Randomized Search demonstrates superior efficiency in terms of computational costs and time, enabling the acquisition of effective and reliable parameter sets within a reduced timeframe (Ibid).

3 RESULT & ANALYSIS

3.1 Purchase Power Prediction

After classifying the SKUs, the paper comes to user prediction. The purchase_power attribute was used as our target variable Y, while X include user_level, plus, gender, age, marital_status, education, and city_level. The dataset was randomly partitioned into a train set and a test set with an 8:2 ratio. Considering the numerical significance of purchase_power, this study posits that it cannot be simply categorized as a classification problem. Instead, user purchasing power should be treated as a continuous variable through regression analysis and subsequently rounded to classify their purchasing power levels. Consequently, this paper introduces Linear Regression model and Decision Tree regression to evaluate their performance using the Mean Squared Error (MSE) metric, with the results presented in Table 2.

A comparison of MSE between training and testing sets reveals that the disparities between training error and testing error for both models are minimal, suggesting the absence of pronounced overfitting or underfitting. The MSE for Decision Tree Regression is marginally lower than that of Linear Regression across both training and testing datasets. Hence, it can be inferred that Decision Tree Regression exhibits a slight advantage and robust generalization capability. For further investigations into user purchasing power, Decision Tree Regression may be considered as a preferable approach.

3.2 SKU Preference Prediction

In the context of clustering SKU categories, user click data across various product categories was quantified. Consequently, user categories were segmented and assigned specific category codes, denoted as Y. Additionally, variables including plus, gender, age, marital_status, education, and city_level were selected as independent variables X to model and predict user preferences. The dataset was still randomly partitioned into a train set and a test set with an 8:2 ratio.

3.2.1 No-Boundary Classification. Unconstrained by boundaries, users were categorized into four distinct groups based on their click behavior across four SKU categories. Users with a higher number of clicks on luxury products compared to cost-effective products and bargains were classified as "High luxury preference users", assigned a category code of 0. Users exhibiting a greater number of clicks on cost-effective products over luxuries and bargains were categorized as "Value-for-money preference users", with a category code of 1. Users who predominantly click on bargains more than on luxury and cost-effective products were classified as "Low-price preference users", assigned a category code of 2. All other scenarios were collectively referred to as other preference users, designated with a category code of 3.

The study employed Decision Tree Classifier, Random Forest Classifier, and Stacking model to predict user preferences. Parameter optimization was attempted using Randomized Search CV. Table 3 shows the accuracy comparison of three models before and after parameter tuning and feature engineering. Figure 3 shows the ROC curve of the Decision Tree model after RFE and parameter tuning considering its best model performance.

Based on the accuracy outcomes presented in Table 3, it is evident that parameter tuning and feature engineering substantially enhanced the model's accuracy. The application of RFE to discard non-essential features rendered the model more concise and efficient. Furthermore, the implementation of Randomized Search for

Table 3: Comparison of model accuracy without considering the boundary problem.

Classification method	Original		After RFE & Parameter tuning	
	Train Accuracy	Test Accuracy	Train Accuracy	Test Accuracy
Decision Tree Classifier	0.615	0.539	0.587	0.567
Random Forest Classifier	0.612	0.552	0.608	0.558
Stacking	0.588	0.567	0.587	0.566

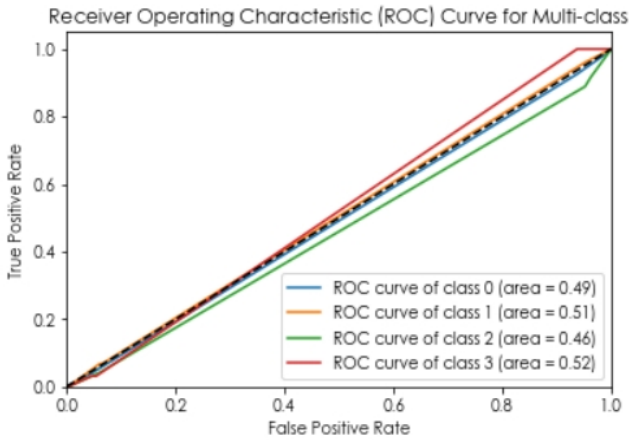

Figure 3: ROC curve of the Decision Tree model.

parameter optimization further augmented the model's predictive performance. Conversely, the Stacking model did not yield a significant improvement, potentially attributable to the limited prediction variance among the base models, thereby hindering the combined model's ability to fully exploit the strengths of each individual base model. Additionally, the effectiveness of Stacking models may be constrained by the size and heterogeneity of the training dataset. The Decision Tree model emerged as the superior performer in this context, likely due to its capability to manage nonlinear relationships and its adaptability to the scale and dimensionality of the data. Decision Trees are inherently interpretable and visually representable, which facilitates the elucidation of the influence of various features on the outcomes, making them particularly valuable in the analysis of user behavior.

3.2.2 Boundary-Consideration Classification. With boundary issues in mind, the article removed the "Other" category and merged the three SKU categories into two results: Effective + Bargain and Luxury + Effective. To explain, the clicking behavior of users may span multiple categories concurrently. For instance, a user could have an interest in both luxury goods and low-priced products. Thus, it is imperative to adopt a methodology that accommodates such scenarios, thereby making the classification more robust and realistic. Relying solely on a singular criterion for categorization may result in misclassification. For example, a user who marginally clicks more on luxury goods but predominantly engages with discounted products may be inaccurately classified as interested primarily in luxury goods. Addressing the issue of fuzzy boundaries, this research investigates the utilization of comprehensive indicators to classify users, aiming to ensure stability in classification results and prevent significant category shifts due to minor variances in click patterns. To this end, the study introduces a boundary blurring concept based on the click-through ratio. By aggregating data, it calculates the total number of user clicks across three categories, excluding the "Other" category. The paper determined the proportion of clicks on luxury and cost-effective products relative to the total number of clicks as *luxury_effective_ratio*. Additionally, calculating the ratio of the total clicks on cost-effective and bargains to the overall clicks as *effective_bargain_ratio*. A threshold of 0.2 was employed to assess the absolute difference between these two ratios. If the difference exceeds this threshold, the user exhibits definitive characteristics corresponding to their preliminary classification category. For users who meet this threshold, a secondary classification is performed based on the magnitudes of the *luxury_effective_ratio* and the *efficient_bargain_ratio*. If the *luxury_effective_ratio* surpasses the *efficient_bargain_ratio*, the user would be classified as 1, indicating a greater inclination towards luxury goods and cost-effectiveness. Conversely, if the *luxury_effective_ratio* is lesser, the user would be classified as 0, indicating a preference for cost-effectiveness and bargains.

Considering that the model becomes a binary classification problem after merging user categories, this paper uses Logistic regression, Decision Tree, and Random Forest models for prediction. Feature engineering and parameter tuning were also applied to test their effectiveness in improving model performance. The outcomes of the predictive models are detailed in Table 4. Figure 4 shows the ROC curve of the Random Forest model after RFE and parameter tuning, which shows the best model performance.

The comparison of the model results presented in Table 4 indicates that the Logistic Regression model is relatively simple, exhibiting minimal differences between training and testing accuracy. This suggests stability, though the performance of the model is moderate. The initial Decision Tree Classifier model exhibited overfitting, which was mitigated to a certain extent following parameter tuning. Although the testing accuracy showed slight improvement, the overall performance remained relatively average. Conversely, parameter tuning for the Random Forest Classifier resulted in a substantial enhancement of test set accuracy, demonstrating superior performance. This suggests that the Random Forest model possesses strong generalization ability and robustness, making it the most effective for user preference classification tasks.

On the other hand, considering the information in both Table 3 & 4 and Figure 3 & 4, the study notes that considering of the boundary issue is conducive to the result, with both amelioration in both AUC size and accuracy, improving from an average of 0.50 & 0.57

Table 4: Comparison of model accuracy considering the boundary problem.

Classification method	Original		After RFE & Parameter tuning	
	Train Accuracy	Test Accuracy	Train Accuracy	Test Accuracy
Logistic Regression	0.725	0.737	0.728	0.737
Decision Tree Classifier	0.750	0.733	0.734	0.735
Random Forest Classifier	0.749	0.728	0.738	0.747

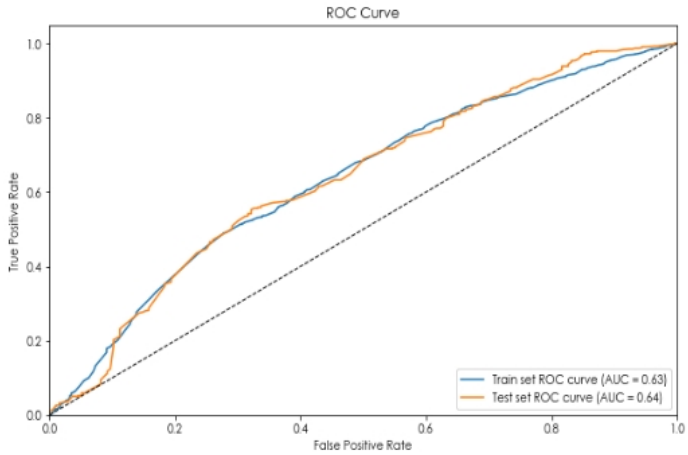

Figure 4: ROC curve of the Random Forest model.

to about 0.64 & 0.74. Overall, it is imperative to account for boundary ambiguity when engaging in user classification and prediction tasks. User behavior and preferences are often not absolutely clear and have a certain degree of ambiguity, as individuals may not exhibit unequivocal preferences for specific types of products. Consequently, addressing the issue of boundary blurring allows for a more accurate representation of user behavior and facilitates more nuanced user differentiation. This, in turn, enhances the predictive accuracy of the model and improves the feasibility of developing more precise recommendation systems, particularly in the context of JD's future endeavors.

4 CONCLUSION

This study elucidates critical insights into user preferences through an analysis of purchasing behaviors among JD.com customers, highlighting the essential role of these insights in refining customer segmentation strategies. The findings indicate substantial differences in purchasing preferences across distinct user demographics, particularly with respect to city level and educational background. Specifically, users residing in higher-tier cities exhibit a greater propensity to purchase 1P products, while those with higher educational attainments show a preference for luxury goods. Regarding user preference prediction, classification models that incorporate boundary considerations demonstrate marked superiority over traditional models. The application of a Random Forest classifier in this study yielded a prediction accuracy of 74% on the test set. The efficacy of the model optimization strategy was further corroborated through a comparative analysis of model accuracy before and after feature engineering and parameter tuning.

The practical implications of this study suggest that JD.com can implement more precise marketing strategies and enhance customer satisfaction by gaining a nuanced understanding of the product selection behavior of various user groups. By meticulously segmenting user characteristics, JD.com can refine product recommendations and marketing strategies, thereby bolstering market competitiveness. Furthermore, the study underscores the significance of accounting for boundary blurring phenomenon in classification methods, which markedly enhances prediction accuracy.

In conclusion, this study provides an efficacious approach for JD.com and other e-commerce platforms to augment the prediction accuracy and customer satisfaction of personalized segmentation strategies. This not only enables companies to gain profound user insights but also offers a theoretical foundation for the development of more precise marketing strategies.

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